

Curated bibliography

Type-1, $N = 3$ Landau–Zener closed-form derivation

Reading roadmap for the deep literature review

Purpose. This document is a curated, annotated bibliography for the open closed-form derivation of the Type-1, $N = 3$ ungauged multistate Landau–Zener transition matrix. Phase I delivered two Brundobler–Elser extreme-survival entries exactly. Phase II tested three exact-WKB literature routes and all failed empirically. Phase III narrowed the open problem to: identify a $U(3)$ -valued function $D(\epsilon, \gamma, a)$ that produces the remaining two transcendental entries of P . Each reference below is annotated with what it gives us, what to extract on a careful read, and a priority tier: **P1** (must read first; resolves a Phase II/III gap), **P2** (secondary; framework context), **P3** (supplementary; useful for write-up). A 5-pass reading roadmap appears at the end.

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1 Spectral networks (Gaiotto–Moore–Neitzke)

The foundational framework for the WKB connection matrix as a geometric object on a Riemann surface. The Type-1, $N = 3$ spectral curve $\Sigma_Y : \mu^2 = Q_4 L_H^2 / p^2$ (genus 1, $b_1 = 2$) sits squarely inside this framework.

Gaiotto, Moore, Neitzke. *Spectral networks.* arXiv:1204.4824 (2012). *Annales Henri Poincaré* 14 (2013) 1643–1731.

Relevant sections: §§5–6 (wall structure, BPS counting), §10 (non-abelianisation map).

Annotation: The geometric language of spectral networks: BPS trajectories emanating from turning points, walls of marginal stability, and the non-abelianisation map from non-abelian flat connections to flat $U(1)$ connections on the spectral cover. **What it gives us:** the topological skeleton on which the inter-window connection matrix lives; the precise definition of the ω -pair $K_{\pm}(\gamma) = I + \mu_{\gamma} E_{ij}$, $I + (1/\bar{\mu}_{\gamma}) E_{ji}$. **What to extract:** §5.3 the sliding-flag construction; §6.2 the K -wall factor and its reciprocal- μ pair; §10.2 the abelianisation map; §10.3 the ordered chamber product.

Priority: P1

Gaiotto, Moore, Neitzke. *Spectral networks and snakes.* arXiv:1209.0866 (2012). *Annales Henri Poincaré* 15 (2014) 61–141.

Relevant sections: §3 (abelianisation map on snake diagrams).

Annotation: Cluster-coordinate refinement of GMN 2012. Snake diagrams are the combinatorial encoding of the spectral network on a punctured surface useful for computing the connection matrix entry-by-entry. **What it gives us:** a concrete worked example of non-abelianisation in low rank with explicit formulas. **What to extract:** §3.2 the cluster-coordinate decomposition of the connection matrix; relate to Iwaki–Nakanishi (§3 below).

Priority: P2

2 Non-abelianisation and Hitchin systems (Hollands–Neitzke)

The direct reference family for the project’s setting. Hollands–Neitzke extend GMN to explicit connection-matrix formulas for rank-2 and rank-3 Hitchin systems, including the genus-1 case that matches our Σ_Y exactly.

Hollands, Neitzke. *Spectral networks and Fenchel–Nielsen coordinates.* arXiv:1312.2979 / 1402.1131 (2016). *Lett. Math. Phys.* 106 (2016) 811–877.

Relevant sections: §§4 (explicit rank-2 abelianisation), §4.4 (the ω -pair to N^{proper} collapse).

Annotation: The proof we need most: that for a rank-2 spectral network with the canonical BPS-ray choice, the inner -pair product $K_{-}(\gamma) \cdot \Phi_{\text{mid}} \cdot K_{+}(\gamma)$ collapses *exactly* to the Joye–Kunz–Pfister Stokes-phased N^{proper} block. **What it gives us:** the algebraic identity that ratifies the expected Route A closed form $U_{HN} = \Phi_R N_B^{\text{proper}} \Phi_{AB} N_A^{\text{proper}} \Phi_L$. **What to extract:** §4.4 Lemma giving the collapse; §4.2 the explicit $SL(2, \mathbb{C})$ -valued formula.

Priority: P1

Hollands, Neitzke. *BPS states in spectral networks of arbitrary rank.* arXiv:1611.04317 (2016).

Relevant sections: §2 (rank- N generalisation), §5 (genus- ≥ 1 examples).

Annotation: Rank-3 / higher-rank generalisation that includes our $N = 3$ case without reduction to rank 2. **What it gives us:** confirmation that the rank-3 -pair construction works on genus-1 Hitchin systems and that the BPS-state count is fixed by b_1 of the cover. **What to extract:** §2.3 the rank- N wall-jump formula; the worked $SL(3)$ example for spectral curves with $b_1 = 2$.

Priority: P2

Hollands, Neitzke. *Exact WKB and abelianization for the T_3 equation.* arXiv:1906.04271 (2019). *Commun. Math. Phys.* **388** (2021) 1571.

Relevant sections: §§5–6 (chamber-structure proof for genus-1 covers with $b_1 = 2$).

Annotation: **The direct genus-1 reference for our setting.** The T_3 equation is a third-order linear ODE whose spectral cover has exactly the topology of our Σ_Y (genus 1, $b_1 = 2$). **What it gives us:** the proof that the GMN ordered product at the canonical phase collapses to two N^{proper} factors plus a diagonal inter-window factor Φ_{AB} , with no extra inter-window BPS state in the relevant chamber. This is the load-bearing theorem for Route A’s expected closed form. **What to extract:** §5 chamber-structure analysis; §6 the final connection-matrix formula in the relevant chamber; convention table for θ -phase and -pair side choice.

Priority: P1

3 Iwaki–Nakanishi cluster algebras

The cluster-algebra dual of GMN: Voros symbols become cluster Y -variables and Stokes phenomena become cluster mutations. Provides an alternative vocabulary for the same connection matrix.

Iwaki, Nakanishi. *Exact WKB analysis and cluster algebras.* arXiv:1401.7094 (2014). *J. Phys. A* **47** (2014) 474009.

Relevant sections: §§3–4 (cluster mutation as Stokes phenomenon), §5 (the connection matrix formula as mutation product).

Annotation: The cluster-algebraic vocabulary for exact WKB. Each mutation contributes a Faddeev quantum dilogarithm Φ_b factor; the connection matrix is a finite product of these along a cluster path. **What it gives us:** Route B’s framework; explicit formula for the mutation product that produces a $U(N)$ connection matrix from $N + 1$ cluster variables. **What to extract:** §3.2 mutation rule; §4 the T -system / Y -system identity that our Sinitzyn–Chernyak Track 3 result corresponds to.

Priority: P1

Iwaki, Nakanishi. *Voros coefficients of the Schrödinger equation.* arXiv:1409.4641 (2014). *J. Phys. A* **47** (2014) 474012.

Relevant sections: §§2–3 (Voros symbols as cluster coordinates).

Annotation: Companion paper to 1401.7094, focused on the Schrödinger pencil $\hbar^2 \psi'' + Q(u)\psi = 0$. **What it gives us:** explicit Voros-symbol formulas for the rank-2 reduction of our quotient-ring scalar equation. **What to extract:** the formula relating $\oint \sqrt{Q} du$ to the cluster Y -variable, and the Faddeev Φ_b argument at finite b .

Priority: P2

4 Quantum dilogarithm (Faddeev)

The fundamental special function of the cluster-mutation route. Plays the same role for cluster algebras that Γ plays for the Joye–Kunz–Pfister Stokes block.

Faddeev. *Discrete Heisenberg–Weyl group and modular group.* *Lett. Math. Phys.* **34** (1995) **249–254**.

Relevant sections: Eq. (5) (the Φ_b definition); §2 (the five-term relation).

Annotation: Original definition of the non-compact / modular quantum dilogarithm $\Phi_b(z)$ as a contour integral or infinite product. **What it gives us:** the function our finite- b Faddeev refactor needs. **What to extract:** the integral representation (better behaved than the infinite product for large $|z|$); the $b \rightarrow b^{-1}$ duality.

Priority: P1

Faddeev, Kashaev. *Quantum dilogarithm.* *Mod. Phys. Lett. A* **9** (1994) **427–434**.

Relevant sections: §§2–3 (basic properties; semiclassical limit).

Annotation: The original integral representation of Φ_b . **What it gives us:** the $b \rightarrow 0$ classical limit (recovers Li_2); the connection to Γ -functions in the appropriate limit. **What to extract:** the precise small- b asymptotic expansion.

Priority: P2

5 Aoki–Kawai–Takei exact WKB / DDP

The framework closest to Phase II’s Route C: discontinuity formulas for exact-WKB connection matrices at simple and coalescing turning points.

Aoki, Kawai, Takei (and collaborators). *Exact WKB analysis* (review). E.g. RIMS preprint series and Kyoto lecture notes, **1992–2010**.

Relevant sections: §§4–5 (DDP at simple and coalescing turning points; Whittaker / Bessel matching).

Annotation: The exact-WKB connection-matrix programme. **What it gives us:** the proper formula for the connection matrix at two coalescing turning points (modified-Bessel index $1/3$ at argument $2|V|/3$). **What to extract:** the Wronskian-normalised form of the Bessel block (Phase III prototype used a heuristic $\sin(\arg V)$ which was wrong); the precise sheet selection for the inter-window Voros symbol.

Priority: P1

Delabaere, Dillinger, Pham. *Résurgence de Voros et périodes des courbes hyperelliptiques.* *Ann. Inst. Fourier* **43** (1993) **163–199**.

Relevant sections: Theorems II.2, II.3 (the discontinuity formula at a simple turning point).

Annotation: The original DDP discontinuity formula: the analytic continuation of a WKB symbol around a turning point picks up a controlled Γ -function-valued jump. **What it gives us:** the elementary connection block we already implement in `assay.ddp.j_ddp`; the framework that AKT extends to coalescing turning points. **What to extract:** Theorem II.3 (the universal form of the discontinuity); the residue computation at the turning point.

Priority: P1

6 Multistate Landau–Zener – historical core

Brundobler, Elser. *S-matrix of multistate Landau–Zener problem.* J. Phys. A **26** (1993) 1211–1227.

Relevant sections: Throughout; especially the extreme-level survival formula.

Annotation: The original derivation of the Brundobler–Elser product: $P[\text{extreme level survives}] = \prod q_{ij}$ with $q_{ij} = \exp(-2\pi\Gamma_{ij})$. **What it gives us:** the two exact-elementary entries of the Type-1 $N = 3$ matrix P (Phase I deliverable). **What to extract:** the topological argument that fixes the two extreme corners independently of all integrability data; the implicit “no-self-interaction” assumption.

Priority: P1

Demkov, Osherov. *Stationary and nonstationary problems in quantum mechanics that can be solved by the method of factorization.* Sov. Phys. JETP **24** (1967) 916.

Relevant sections: Sections on the equal-slope special case.

Annotation: The original three-state LZ with one “selector” level crossing the other two. **What it gives us:** historical context for the Type-1 geometry (Demkov–Osherov is the symmetric special case; Type-1 is the generic version). **What to extract:** the sequential-LZ-product structure that DOES hold in the Demkov–Osherov special case; the precise extension to the generic Type-1 is what Phase III is solving.

Priority: P2

Sinitsyn. *Multistate Landau–Zener problem.* arXiv:cond-mat/0212017 (2002) and successors; recent review in New J. Phys. **19** (2017) 075008 with Chernyak.

Relevant sections: §§2–3 (integrability conditions, hidden symmetry).

Annotation: The integrable hierarchy of multistate LZ, including the Sinitsyn–Chernyak commuting-triple condition $[H, H'] = 0$. **What it gives us:** Track 3 verified this commuting structure to 7×10^{-15} ; the hierarchy pins the two extreme diagonals but leaves the middle survival open. **What to extract:** the identification of which P entries are integrability-fixed vs. genuinely transcendental — and exactly how many free parameters remain.

Priority: P1

7 Two-level LZ with Stokes-phase corrections

Joye, Kunz, Pfister. *Exponential estimates in adiabatic quantum evolution.* Ann. Inst. H. Poincaré **54** (1991) 343–378. Companion: J. Math. Phys. **32** (1991) 3328.

Relevant sections: §§2–3 (the rigorous $\varphi_S(\delta)$ formula).

Annotation: The mathematically rigorous derivation of the Stokes phase $\varphi_S(\delta) = \delta(\log \delta - 1) + \pi/4 + \arg \Gamma(1 - i\delta)$ for the two-level LZ amplitude block. **What it gives us:** the function we implement in `assay.lz_stokes_phased`; the closed-form expression we expect to see emerge from the GMN -pair construction. **What to extract:** the proof that this is the unique non-perturbative correction to the elementary BE amplitude in the 2-level case; the convention for the sign of φ_S .

Priority: P1

Berry. *Histories of adiabatic quantum transitions.* *Proc. R. Soc. A* **429** (1990) 61–72.

Relevant sections: §§3–4 (Stokes lines and the histories interpretation).

Annotation: Berry’s “matching” interpretation of the LZ Stokes phenomenon. **What it gives us:** physical context for φ_S as a matching condition across Stokes lines, useful for explaining the result. **What to extract:** the geometric picture of the Stokes lines in the complex time plane and how they connect to the GMN spectral network.

Priority: P2

8 Hitchin integrable systems (foundational)

Hitchin. *Stable bundles and integrable systems.* *Duke Math. J.* **54** (1987) 91–114.

Relevant sections: §§4–6 (definition; rank- N case; genus-1 special case).

Annotation: The foundational paper defining Hitchin systems. **What it gives us:** the rigorous identification of our commuting triple (H_x, Y, R) from Track 3 as a Hitchin system on a punctured genus-0 base with rank-3 spectral cover (which IS our Σ_Y with $b_1 = 2$). **What to extract:** §6 the genus-1 case; the symplectic structure that the Hollands–Neitzke abelianisation preserves.

Priority: P2

Hitchin. *The self-duality equations on a Riemann surface.* *Proc. London Math. Soc.* **55** (1987) 59–126.

Relevant sections: §§5–6 (rank- N moduli space; the integrable hierarchy).

Annotation: Companion paper to Duke 1987; constructs the full Hitchin moduli space. **What it gives us:** the moduli-space perspective on why our two-parameter open content sits in a 6-dimensional $U(3)$ chamber. **What to extract:** the parameter count for the rank-3 / genus-1 Hitchin moduli (matches Phase III’s 6-after-gauge count of D).

Priority: P3

9 Cluster coordinates (Fock–Goncharov, Allegretti)

Fock, Goncharov. *Cluster ensembles, quantization and the dilogarithm.* arXiv:math/0311149 (2003). *Ann. Sci. É.N.S.* **42** (2009) 865.

Relevant sections: §§2–3 (cluster ensembles; X - and A -coordinates).

Annotation: The general framework of cluster ensembles. **What it gives us:** the rigorous structure that contains Iwaki–Nakanishi as a special case; the duality between Voros symbols (X -coordinates) and shear coordinates (A -coordinates). **What to extract:** the mutation rules and the dilogarithm identities that constrain the cluster product.

Priority: P2

Allegretti. *Voros symbols as cluster coordinates*. arXiv:1810.09433 (2018). Int. Math. Res. Not. (2019).

Relevant sections: Throughout; especially the cluster-coordinate identification.

Annotation: Explicit identification of WKB Voros symbols with Fock–Goncharov cluster coordinates.

What it gives us: the bridge that makes Iwaki–Nakanishi rigorous; the proof that the cluster Y -variable equals $\exp(\oint \sqrt{Q} du)$ along the appropriate cycle. **What to extract:** the precise dictionary between WKB and cluster vocabulary for our genus-1 case.

Priority: P1

10 Special functions and reference handbooks

Olver, Lozier, Boisvert, Clark (eds.). *NIST Digital Library of Mathematical Functions*. dlmf.nist.gov, ongoing. Companion: Abramowitz & Stegun, *Handbook of Mathematical Functions* (1964), Ch. 9–10.

Relevant sections: Ch. 9–10 (modified Bessel functions); Ch. 5–6 (Γ and related functions).

Annotation: Reference for the modified Bessel functions I_ν, K_ν at $\nu = 1/3$ needed for the AKT block, and for the Γ -function identities used throughout. **What it gives us:** numerical values, asymptotic expansions, Wronskian relations. **What to extract:** the Wronskian $I_\nu K'_\nu - I'_\nu K_\nu = -1/z$ (DLMF 10.28.2) which fixes the unitary normalisation of the AKT block.

Priority: P3

Reading roadmap (5 passes)

The references above are organised by framework. The order below is the recommended *reading* order, driven by which Phase II/III gap each pass closes. For each item the annotation tells you exactly what to extract — this is not an open-ended read.

Pass 1 – Route A gap fix (1–2 days)

Goal. Extract the -pair side-tracking convention that fixes the Phase III Route A prototype’s catastrophic overflow (residual 10^{23} to 10^{193} from naive $I + \mu E_{ij}$ insertions with $|\mu| \sim e^{50}$).

Read.

- Hollands–Neitzke arXiv:1906.04271 §§5–6 (the genus-1 chamber-structure proof for T_3 , matching our Σ_Y with $b_1 = 2$).
- Gaiotto–Moore–Neitzke arXiv:1204.4824 §5 (the $K_{\pm}(\gamma)$ pair definition) and §10.3 (the ordered chamber product).

Extract. The exact algebraic identity $K_{-}(\gamma) \cdot \Phi_{\text{mid}} \cdot K_{+}(\gamma) = N_X^{\text{proper}}$ for an inner -pair, where N_X^{proper} is the Joye–Kunz–Pfister Stokes-phased block. This is the load-bearing identity for Route A’s expected closed form $U_{HN} = \Phi_R \cdot N_B^{\text{proper}} \cdot \Phi_{AB} \cdot N_A^{\text{proper}} \cdot \Phi_L$.

Pass 2 – Route B gap fix (2–3 days)

Goal. Extract the log-space cluster-mutation product and the finite- b Faddeev kernel that would fix Phase II Route B’s Voros- Y overflow.

Read.

- Iwaki–Nakanishi arXiv:1401.7094 §§3–4 (the mutation rule and the cluster product).
- Faddeev Lett. Math. Phys. 34 (1995) eq. (5) and §2 (the Φ_b definition and five-term relation).

Extract. The explicit mutation formula $\mu_k(Y_j) = Y_j(1 + Y_k^{\mp 1})^{B_{jk}}$ with the Φ_b contribution per mutation; the convention for finite b at our $\epsilon = 1$ semiclassical limit. (Route B was not implemented in Phase III but is architecturally sufficient if the Phase II combinatorics is corrected.)

Pass 3 – Route C gap fix (2–3 days)

Goal. Extract the correct inter-window Voros symbol (sheet selection) and the Bessel-index-1/3 connection block. The Phase III Route C prototype used a heuristic $\sin(\arg V)$ for the mixing magnitude and integrated on the wrong sheet ($V_{\text{new}} = 2V_{\text{old}}$ across all samples).

Read.

- AKT review (Kyoto / RIMS lecture notes) §§4–5.
- Delabaere–Dillinger–Pham Ann. Inst. Fourier 43 (1993) §III (the discontinuity formula).

Extract. The Wronskian-normalised Bessel block at index $\nu = 1/3$ and argument $2|V|/3$; the precise sheet on which V_{AB} must be integrated (almost certainly the spectator sheet on which the two windows “join,” not the exterior sheet λ_0 the Phase III prototype used).

Pass 4 – Context for write-up (1 day)

Goal. Frame the eventual result in the established LZ literature.

Read.

- Brundobler–Elser J. Phys. A 26 (1993) 1211.
- Joye–Kunz–Pfister Ann. IHP 54 (1991).
- Berry Proc. R. Soc. A 429 (1990) 61.

Extract. The standard convention for the LZ Stokes phase sign; the matching condition across the Stokes lines (for the physical interpretation in the final brief).

Pass 5 – Depth (ongoing)

Goal. Ensure rigorous embedding of the result in the cluster/Hitchin framework for the eventual paper.

Read.

- Allegretti arXiv:1810.09433 (Voros–cluster dictionary).
- Fock–Goncharov arXiv:math/0311149 (cluster ensembles).
- Hitchin Duke Math. J. 54 (1987) 91 (foundational).

Extract. The mapping between our Type-1 $N = 3$ system and the standard Hitchin moduli; the parameter count (6-after-gauge $U(3)$) as a Hitchin-moduli statement.

Total Phase III literature budget: 10 days of focused reading. Pass 1 alone (2 days) should suffice to unblock Route A’s -pair implementation. Passes 2–3 are insurance against Route A failure. Passes 4–5 are write-up depth.

Cross-reference index

For navigation, here is the mapping between Phase II/III gaps and the references that close them:

Gap	Primary reference	Secondary
Route A -pair side-tracking	Hollands–Neitzke 1906.04271 §§5–6	GMN 1204.4824 §5
Route A genus-1 collapse to N_{proper}	Hollands–Neitzke 1402.1131 §4.4	—
Route B log-space mutation	Iwaki–Nakanishi 1401.7094 §3	Faddeev LMP 34 (1995)
Route B finite- b Faddeev kernel	Faddeev LMP 34 (1995)	Faddeev–Kashaev MPL A 9
Route C Bessel block normalisation	AKT review §5	DLMF 10.28
Route C inter-window V_{AB} sheet	AKT review §4	DDP 1993
Stokes phase $\varphi_S(\delta)$	Joye–Kunz–Pfister 1991	Berry 1990
BE extreme-survival formula	Brundobler–Elser 1993	—
Integrability / commuting triple	Sinitsyn–Chernyak NJP 19 (2017)	Sinitsyn 2002
Cluster coordinates	Allegretti 2018	Fock–Goncharov 2003
Hitchin moduli context	Hitchin Duke 1987	Hitchin LMS 1987

Project context references (already shipped)

For self-consistency, the Phase II/III status is documented in:

- `window_period_BE_open_problem_brief.pdf` — the original open-problem statement.
- `six_track_synthesis_brief.pdf` — Phase I synthesis (Tracks 1–6).
- `phase2_findings_brief.pdf` (11 pages) — the three-route Phase II programme and validation failure analysis.
- `phase3_status_addendum.pdf` (4 pages) — the Phase III Route D negative finding (U(3) richness intrinsic).
- `notes/track{1..6}_*.pdf` — per-track Phase I notes.

End of curated bibliography. Total references: 18 across 9 frameworks. Primary reading: 4 P1 references covering Routes A and C plus the Stokes-phase foundation.